



Vertically Offset Overlapping Propellers in Tandem

Jielong Cai¹ and Sidaard Gunasekaran²

University of Dayton, Dayton, OH, 45469

Michael OL³

Folderol, LLC

Modern VTOL configurations often seek minimal footprint simultaneously with minimal disk loading (often competing constraints), the latter to improve propulsive efficiency at low speeds. One approach is to stagger neighboring propellers in partial overlap. We consider such partial overlap of two propellers in tandem, varying the vertical and horizontal spacing, inclination of the propeller plane with respect to the freestream, and advance ratio. The rear propeller was mounted above the plane of the front propeller with a maximum of 25%-diameter overlap. Thrust and torque were measured separately on each propeller. The efficiency of the propeller pair, expressed as a ratio of thrust to power, never completely approaches that of propellers in isolation. But for vertical separation of 0.5 diameter or greater, there is a saturation, where larger separation does not result in further improvement.

I. Nomenclature

C_T	= Propeller Thrust coefficient; $C_T = \frac{T}{\rho n^2 D^4}$	P	= Power, (W)
C_P	= Power Coefficient; $C_P = 2\pi C_Q = \frac{P}{\rho n^3 D^5}$	P_s	= Single Propeller Power Required, (W)
C_Q	= Propeller Torque Coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$	P_t	= Tandem Propeller Power Required, (W)
C_{Qx}	= Propeller Pitching Moment Coefficient	γ	= Propeller Pitch, (m)
C_{Qy}	= Propeller Rolling Moment Coefficient	Q	= Torque, (Nm)
D	= Propeller Diameter, (m)	R	= Propeller Radius, (m)
d	= Distance Between Propeller Hubs, (m)	T	= Thrust, (N)
J	= Propeller Advance Ratio; $J = \frac{V_\infty}{nD}$	T_s	= Single Propeller Thrust, (N)
n	= Propeller Rotational Speed, (RPS)	T_t	= Tandem Propeller Thrust, (N)
		V_∞	= Freestream Velocity, (m/s)
		ρ	= Air Density, (kg/m ³)
		θ	= Propeller-plane Incidence Angle, (deg)

II. Introduction

The increasing usage of propeller-driven aerial vehicles, including unmanned aerial vehicles (UAVs) and eVTOL air taxis, broadens the classical helicopter problem of edgewise flight, where the rotors producing the vehicle lifting force are subjected to a freestream nearly parallel to the rotor-plane. To increase the payload and range, which is to say to reduce the disk loading, one could increase the propeller diameter or increase the number of propellers. The modern fashion is to eschew a small number of large rotors, in particular dispensing with the

¹ Graduate student, Mechanical and Aerospace Department, AIAA Member

² Assistant professor, Mechanical and Aerospace Department, AIAA Member

³ AIAA Associate Fellow

classical single-rotor helicopter, and instead using a panoply of smaller rotors. An additional common constraint is to minimize the vehicle's overall size or ground footprint, to allow for ease of operation in confined areas. One contender to minimize the footprint is coaxial rotors, which are occasionally seen on helicopters to minimize torque imbalance [1]. However, an average performance loss of 15 ~ 25% was reported when compared to a single rotor in hover, as the lower rotor is constantly operating in the wake of the upper rotor [2-5]. This loss is reduced to 9 ~ 15% in high-speed forward flight [2,6,7].

An alternative to a coaxial rotor is to mount the propellers at close proximity in the same plane, for example as in the VoloCity air taxi [8] in Figure 1. The loss in hover performance is negligible when placing the propellers over $1.1 d/D$ apart from the hub of the rotor [5, 10–12, 19]. However, an 11 ~ 18% performance loss in low-speed forward flight is found at $d/D < 3$ for tandem rotors, due to the rotor interaction in forward flight adversely affecting the rear rotor [2, 6, 7, 11-14, 17]. Moreover, an overall thrust difference between the front and rear rotor can be up to ~40% based on flight test data in Ref. [13] and Ref. [18].

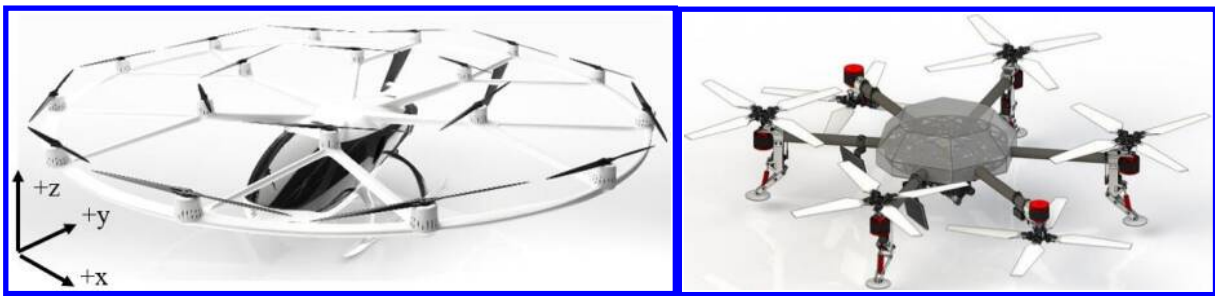


Figure 1. VoloCity air taxi [8] (left) and JAXA Mars exploration helicopter [9] (right).

One method to minimize the performance loss in forward flight is to increase the downstream distance between the propellers to minimize interactions [7, 13, 17]. Ref. [7] and Ref. [13] found that placing the propellers 3 diameters apart ($d/D_f > 3$, Figure 1) results in negligible interference. However, such large spacing vitiates compactness (Figure 1).

An alternative is to place the rotors in tandem, with a vertical displacement, such as the JAXA Mars exploration helicopter [9] (Figure 1). Based on flow visualization studies and force-based testing, Ref. [13] suggests that the reduction of the aft rotor's performance is due to the changes in its inflow angle caused by the front rotor, which agrees with the Fast Multirotor Performance Prediction Model (FMPP) developed by Tsaltas [14]. Ref. [7] also concludes that placing the front rotor below the rear rotor reduces the influence on the rear rotor by the front rotor. Two-dimensional parametric studies (vertical and downstream) on tandem rotor placement in forward flight form the impetus behind the present work

Placing the rear rotor above the front rotor allows for an off-plane partial overlap of the rotor up to $0.5 d_y/D$. The overlapping tandem 2-rotor configuration has been widely studied on helicopters during the development of the CH-47 Chinook [2, 10-12, 15, 16]. For hovering performance, placing the upper rear rotor more than $0.12 d_z/D$ above the lower front rotor results in a minimum impact on the overall performance of both rotors, while overlapping the rotor at $d_y/D > 0.85$ also has a minimum impact on the overall rotor performance. At $d_y/D < 0.85$, the effect of the rotor overlap is almost proportional to the overlapping rotor disk area, which also holds true in forward flight. However, the vertical spacing of the tandem rotor is usually limited to $d_z/D < 0.15$ due to the physical limitations, as a further vertical placement will result in structural disbenefit [5, 13]. In designs shown in Figure 1, the compactness of the propeller placement can be improved if the propellers can be overlapped with a vertical offset. The study focuses on characterizing the performance losses of the two tandem propellers as a function of vertical offset distance and percent overlap.

III. Experimental Setup

A) Data Acquisition

Experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open jet test section. The wind tunnel has an inlet contraction ratio of 16:1, a 762 mm x 762 mm testing section inlet, and a 1.37 m x 1.37 m collector. The freestream range is from 1.5 m/s to 20 m/s, resulting in an advance ratio, J , between 0 and 0.5, with a turbulence intensity of 0.1% measured by a hot wire anemometer.

A sketch of the experimental setup is shown in Figure 2. The propeller is paired with a KDE2814-XF-515 brushless motor and Hobbywing Platinum V4 electric speed controller operating in governor mode for better rotational speed control. The propeller and motor assembly are attached to an ATI Industrial Automation Mini-40 six-component force/torque transducer [20]. The transducer has a maximum of 40N on its x and y-axis, 120N on its z-axis, and maximum torque of 2 Nm for all axes. Thrust is the axial-force along the z-axis of the sensor. The motor torque is the moment of the z-axis, while the propeller pitching moment is the torque on the x-axis, and the propeller rolling moment is the y-axis torque.

The F/T transducer is mounted on a 3D-printed transducer interface, which is then bolted to an 80-20 aluminum extrusion. The extrusion is connected to a Fuyu FSL 40 linear traverse by a 3D-printed traverse interface. The linear traverse is placed at the centerline of the test section and is capable of changing the vertical distance (z) between the propellers. The propellers can be tilted with respect to the tunnel freestream. To change the downstream distance (y), the location of the front rotor is fixed while the rear propeller is moved back or forth. The entire assembly is mounted to a 1.22 x 0.92 x 2.44 meter aluminum frame, secured to the lab floor, the inlet of the wind tunnel, and the ceiling of the wind tunnel plenum.

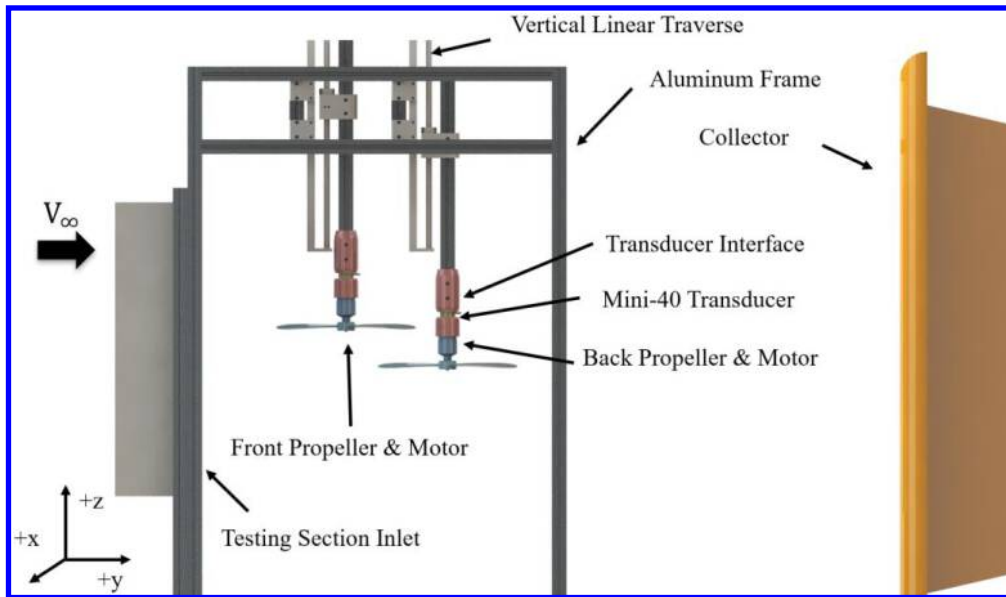


Figure 2. Schematic of the experimental setup at UD-LSWT

The force and torque data sampling rate was 1000 Hz, with two 15-second runs to improve data repeatability for each data point. In an effort to minimize sensor drift, tare values are taken before and after each test, and the average of the two tares is subtracted from the respective thrust and torque readings. A band-pass filter is then applied to filter out any undesired frequency in the signal, including the propeller RPS and its harmonics, and the aluminum frame's natural frequency (~65 Hz). Two 10 mW lasers and two Thorlabs PDA10A Photodiodes are used to measure the rotational speed of each propeller.

B) Test Matrix

The test matrix is in Table 1. The KDE-CF 125 propeller is selected based on its popularity in modern drones. For relevance to actual n-copter drones, the propeller rotational speeds are set such that there is no net pitching moment, or moment about the X-axis in Figure 3; this is akin to trimming a drone in forward flight in pitch [17-18]. As the propeller will generate a pitch-up moment in edgewise flight, based on the result from our previous study [23], the rear propeller must generate a higher thrust to balance the pitch for the lateral stability of the drone. This is shown in the schematic in Figure 3. As the propeller is fixed-pitch, it must operate at a higher rotational speed to achieve trim.

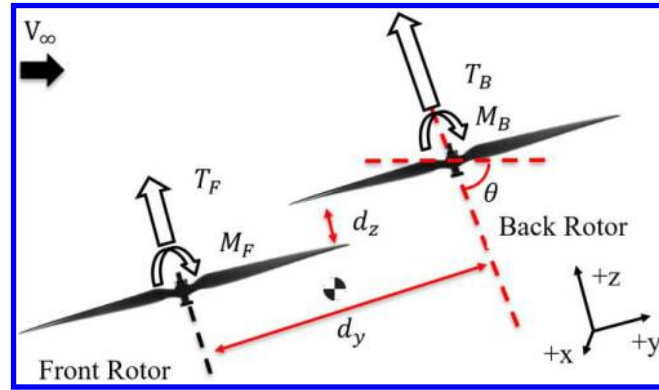


Figure 3 Nomenclature for tandem propellers.

Based on the sensor maximum loading limit and the propeller balance, the front propeller rotational speed is selected at 67 and 75 RPS, resulting in a rear propeller rotational speed ranging between 67 and 115 RPS. The peak Reynolds number is calculated at 75% blade span. To achieve different advance ratios, the propeller RPS is fixed, while the wind tunnel freestream velocity is varied. 75 and 90 degree incidence angles, θ , were chosen, representing realistic operational range for drones in transition to forward flight and in forward flight. The vertical and downstream propeller placement is calculated based on the distance of the propeller hub, shown in Figure 3, while the value of each is between 0 to 0.8 and 0.75 to 1.25, respectively. For the cases where the propeller overlaps ($d_y \leq 1.0$), the vertical distance of $d_z = 0.0$ is not tested due to interference.

Table 1. Test Matrix

Propeller	Rotational Speed (RPS)	Peak Reynolds Number (x1000)	Advance Ratio	Incidence Angle (Deg)	Vertical Placement d_z/D	Downstream Placement d_y/D
KDE-CF125	Front: 67 and 75 Rear: 67 - 115	160	0 - 0.7	75, 90	0, 0.2, 0.4, 0.6, 0.8	0.75, 0.875, 1.0, 1.25

IV. Results

A) Single Propeller

The single propeller is first tested between 45 and 90 degree incidence angles. Three different rotational speeds between 67 and 100 RPS were tested for each propeller to account for the Reynolds number effects. As the difference in the experimental result of thrust and power coefficient between the lowest and highest RPS is less than 4%, the result for the KDE CF125 propeller at 100 RPS is selected and discussed in this section.

The axial thrust coefficient, C_T And power coefficient, C_P , are shown in Figure 4. Results are consistent with our previous study [23] and other studies [24-25], where C_T remains nearly independent of θ or J until $J \sim 0.15$. For $\theta = 75^\circ$, C_T appears to be invariant with J . It increases linearly with J beyond $J \sim 0.15$ when $\theta > 75^\circ$, and decreases linearly with J when $\theta < 75^\circ$. Similarly, C_P remains independent of θ until $J = 0.3$. Beyond $J = 0.3$, the C_P for the different θ cases start to diverge. C_P for $\theta = 90^\circ$ does not follow the trends observed in our previous study, which used propellers designed for fixed-wing-aircraft forward propulsion. There, the C_P at $\theta = 90^\circ$ is smaller than the $\theta = 60^\circ$ and 75° . This is likely due to the fact that the KDE CF125 propeller is designed for edgewise flight, where the blade chord distribution and geometry are different from that of propellers designed for tractor or pusher fixed-wing configurations.

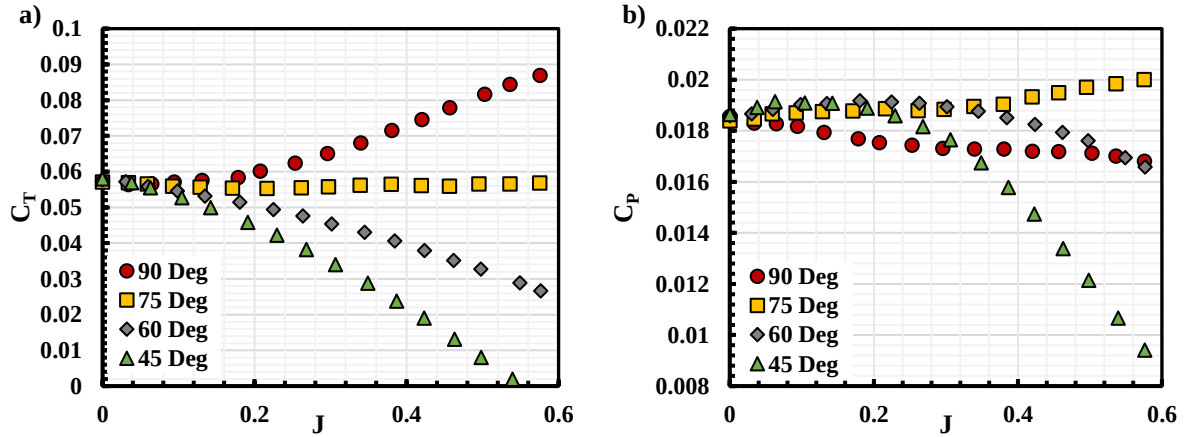


Figure 4. a) C_T and b) C_P for KDE CF-125 propeller vs. J and θ (100 RPS)

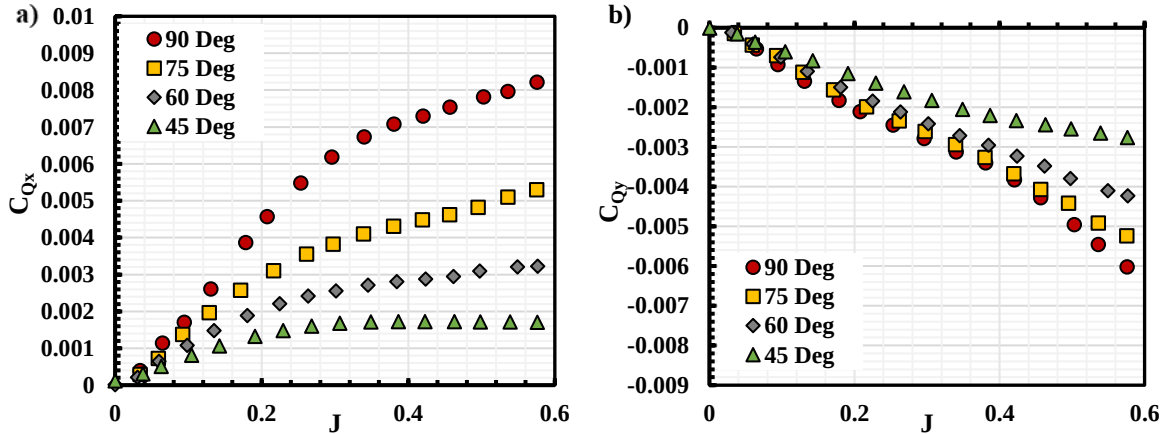


Figure 5. a) C_{Qx} and b) C_{Qy} for KDE CF-125 propeller vs. J and θ (100 RPS).

The propeller pitching moment coefficient, C_{Qx} , and rolling moment coefficient, C_{Qy} , vs. J at different incidence angles are shown in Figure 5. A positive C_{Qx} is pitch-up moment, while a positive C_{Qy} is counterclockwise rolling moment when viewed towards the freestream. C_{Qy} would be the upsetting moment due to advancing/retreating blades, which in a helicopter is canceled with cyclic pitch, but in a drone would require differential rotational speed in the plurality of propellers. Again, the trend is similar to that from our previous study [Ref. 14]. Magnitude of C_{Qx} and C_{Qy} increases with J for all θ cases tested, in roughly comparable amounts, though the pitching moment evinces stronger dependency on θ , with the purely edgewise case as a notable outlier. A noticeable reduction in C_{Qx} curve slope occurs at $J \sim 0.3$, while C_{Qy} increases more or less linearly for all J .

B) Tandem Propeller Configuration at $\theta = 90^\circ$

The tandem propeller configuration is tested under approximate pitch trim configuration, which requires the rear propeller to operate at a higher rotational speed to balance the pitch-up moment generated by the propeller pair. To do this, the rotational speed for the front propeller is fixed, while the rear propeller rotational speed is increased to reach a normalized pitching moment, $C_{Qx\text{ norm}}$, less than 5% for each advance ratio tested. $C_{Qx\text{ norm}}$ is defined as:

$$C_{Qx\text{ norm}} = \frac{C_{Qx\text{ Front}} + C_{Qx\text{ Back}} + (T_{\text{Front}} - T_{\text{Back}}) * 0.5d_y}{C_{Qx\text{ Front}} + C_{Qx\text{ Back}}} \approx 0 \quad (1)$$

Two front propeller rotational speeds are selected, to check the repeatability of the experiment. As the overall Reynolds number effect is relatively small, the result for the front propeller at 4000 RPM (67 RPS) is discussed for all tandem configurations. The rotational speed of the rear propeller with respect to the advance ratio of the front propeller at 67 RPS for $d_y = 1.25$ and 1.0 is shown in Figure 6. From Figure 5, C_{Qx} for a single propeller increases with J , and so in the tandem pair, the rotational speed of the rear propeller needs to increase with J to compensate. there is a knee in the curve for $J \sim 0.4 - 0.5$, consistent with the corresponding trend for a single propeller in Figure 5. By $J > 0.6$, change in rotational speed for the rear propeller becomes negligible. With decrease in d_z , as the influence of the front propeller on the rear propeller increases, the rotational speed of the rear propeller has to increase further to generate the required thrust to achieve trim. This dependency dissipates by $d_z \sim 0.6$ for both d_y values in Figure 6. When reducing d_y , due to the reduction in the moment arm between the propellers, as well as the increase in the propeller interaction, the rear propeller rotational speed increases even further at the same d_z .

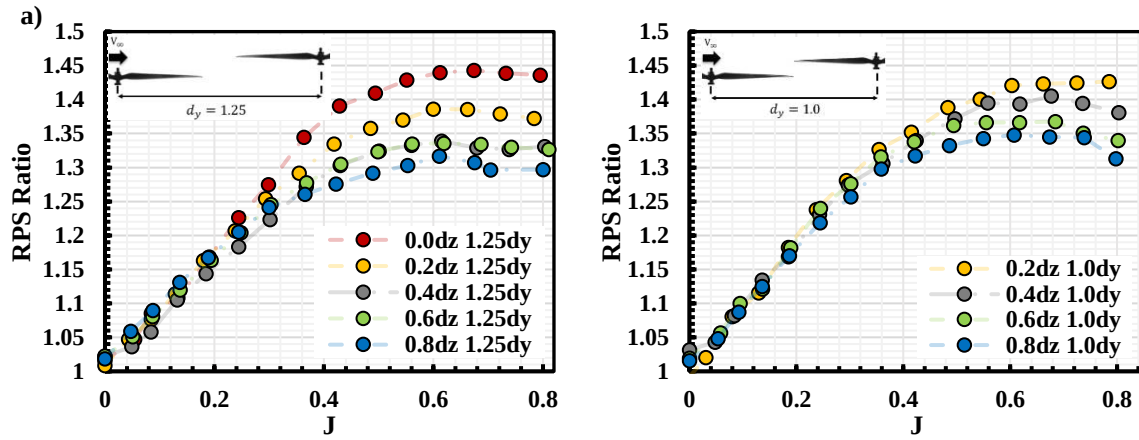


Figure 6. The ratio of rear to front propeller rotational speed at front propeller RPS of 67 and $\theta = 90^\circ$ for a) $d_y = 1.25$ and b) $d_y = 1.0$.

B. 1) Front Propeller

We will first focus on the front propeller and the effect of the rear propeller on the front propeller. The thrust coefficient, C_T at different advance ratios for the add d_y case tested, along with the single propeller which serves as a baseline, are shown in Figure 7. Similar to Figure 4, the change in C_T is nearly constant for $J < 0.2$, and then rises linearly. For $d_y = 1.25$, at largest vertical spacing $d_z = 0.8$, a $\sim 3\%$ increment in C_T is observed across the range of J , as a result of the positive effect of the rear propeller, which is also observed in other studies [13]. This benefit in C_T increases as d_z is reduced. At higher overlap (smaller d_y), this C_T benefit vanishes reduces with the decrement in d_y . For the highest overlap case ($d_y = 0.75$), the C_T benefit at $J > 0.2$ and turns negative at $J < 0.2$ evidently as a result of being mounted in the wake of the rear propeller

[11-12,19]. Reduction in C_T for the front propeller at higher overlap at $J < 0.2$ is also where the propeller wake skew angle is small. Variation across the d_z range also reduces when the propellers overlap.

The power coefficient, C_p , for the cases in Figure 7 is shown in Figure 8. For $d_y = 1.25$, the data for different d_z nearly collapse to one trend, following that of a propeller in isolation. With higher propeller overlap (smaller d_y) variation in C_p vs. d_z increases slightly, with a larger d_z showing a slightly higher C_p .

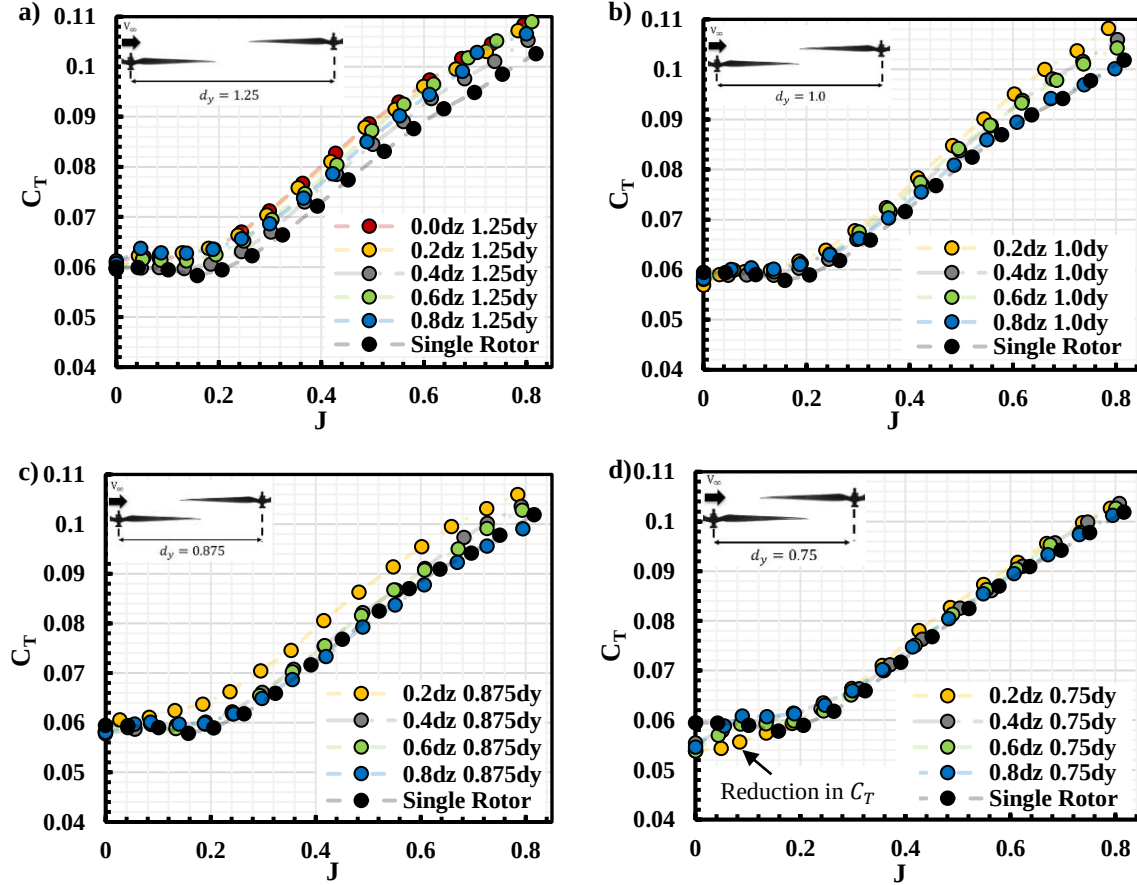


Figure 7. Front Propeller C_T vs. J and d_z , at $\theta = 90^\circ$ for a) $d_y = 1.25$, b) $d_y = 1.0$, c) $d_y = 0.875$ and d) $d_y = 0.75$.

B. 2) Rear Propeller

Since the rear propeller operates at a higher rotational speed than the front propeller, at the same V_∞ , the resulting J for the rear is lower than for the front. Changes in propeller performance due to rotational speed or Reynolds number are less than 3%. We can therefore ascribe changes in propeller performance to the interference between propellers. C_T at different J and d_y is shown in Figure 9. As before, there are no discernible variations for $J < 0.2$. Above $J = 0.2$, there is generally a decrease in C_T , which depends on vertical and streamwise separation, presumably as the rear propeller is at least partially immersed in the front propeller's wake, and this immersion is more pronounced at smaller vertical separation. For $d_y = 1.25$ and $d_y = 1.0$ a d_z of 0.8 nearly restores the solitary-propeller case, while at the smallest d_z the loss in C_T is on the order of 10%-15%. At $d_y = 0.875$ and 0.75, the trend is the same, except that the largest vertical separation appears to evince a slight C_T benefit, or proverse interference. So long as the vertical offset is at least 0.4-0.5, no reduction in C_T is observed, which is to say, that in a multi-propeller drone configuration, a separation of $d_z = 0.4$ is probably sufficient, and effort to space the propellers further apart, is hard to justify from the viewpoint of thrust production.

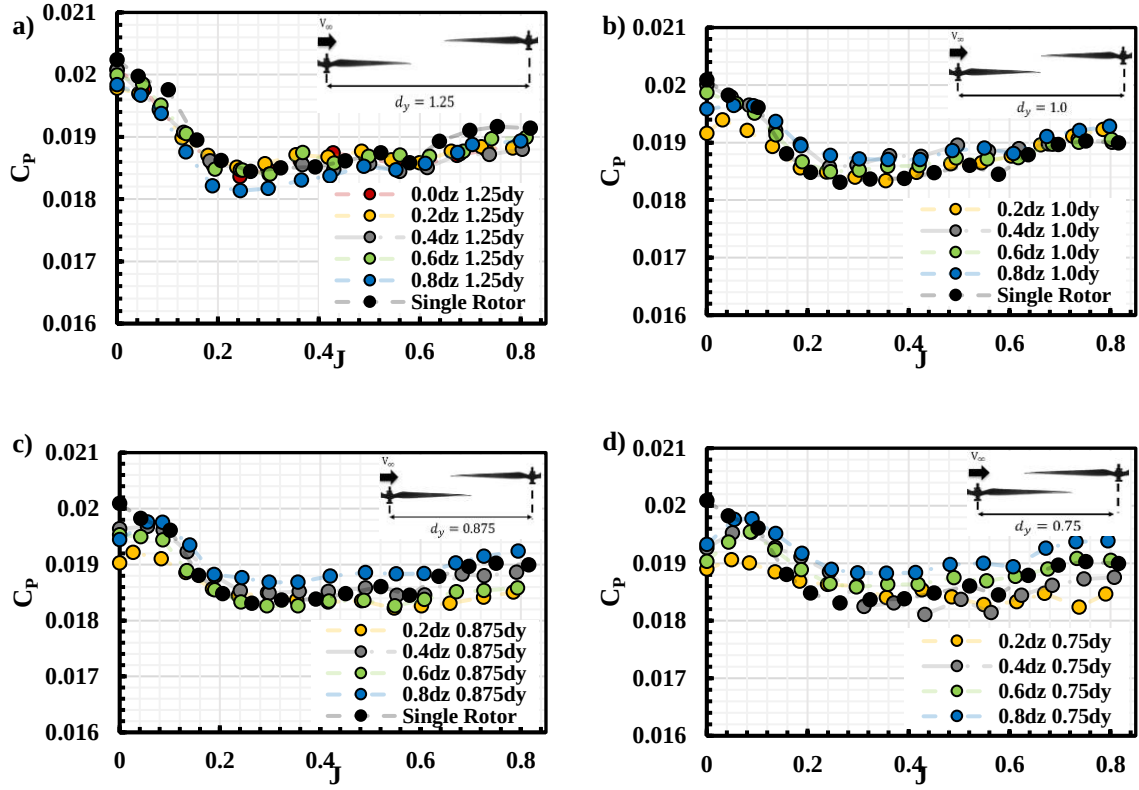


Figure 8. Front Propeller C_p vs. J and d_z at $\theta = 90^\circ$ for a) $d_y = 1.25$, b) $d_y = 1.0$, c) $d_y = 0.875$ and d) $d_y = 0.75$.

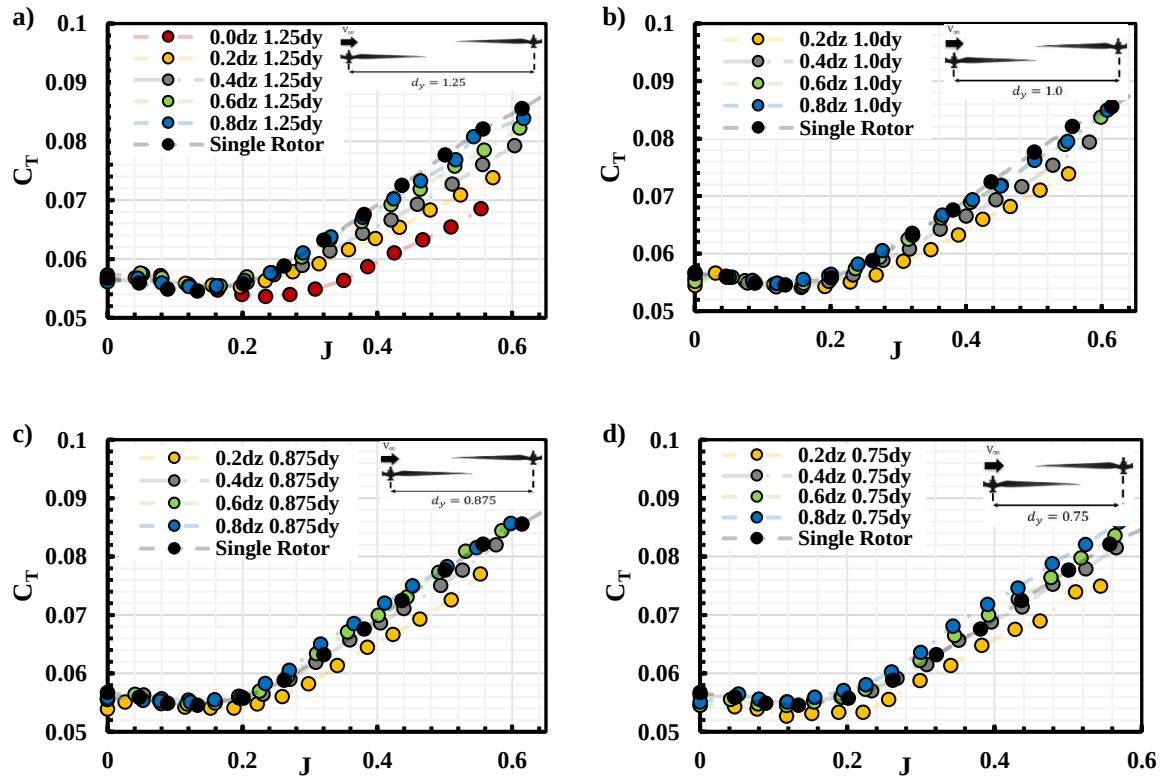


Figure 9. Rear Propeller C_T vs. J at $\theta = 90^\circ$ for a) $d_y = 1.25$, b) $d_y = 1.0$, c) $d_y = 0.875$, and d) $d_y = 0.75$.

At small or no streamwise overlap, the front propeller mainly affects the rear and not vice versa. At high overlap ($d_y = 1.25$), the two propellers are in a partially coaxial orientation. It stands to reason that now the relationship between the two propellers is as it was reversed, such that the rear propeller affects the front one more than the other way around. This depends on advance ratio, as at higher advance ratios the wake of the now-upper (that is, rear) propeller convects downstream, before reaching the plane of the front propeller. Verification would of course require flow visualization.

Figure 10 shows C_P , corresponding to the respective results for the thrust coefficient in Figure 9. For the smallest vertical spacing, C_P is nearly constant with J ; a negative slope in the curve develops as d_z increases. As with the thrust, results for power broadly tend towards the isolated-propeller case, for larger d_z . But unlike for thrust, C_P for larger overlap can get significantly below the single propeller case, for larger d_z .

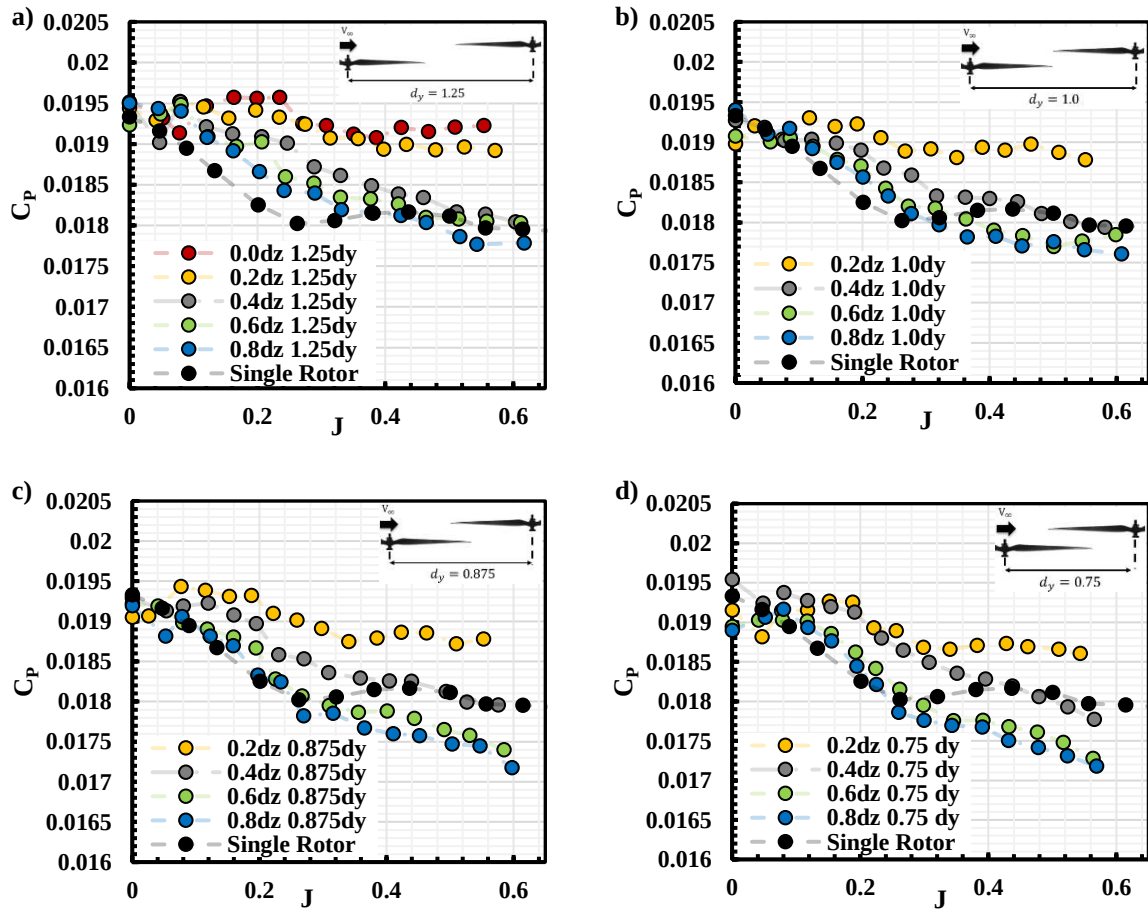


Figure 10. Rear propeller C_P vs. J at $\theta = 90^\circ$ for a) $d_y = 1.25$, b) $d_y = 1.0$, c) $d_y = 0.875$, and d) $d_y = 0.75$

C. Performance Comparisons

What ultimately matters for paired propellers as a system, is not individually thrust or power, but efficiency in thrust production. A ratio of thrust-produced to power-required, at a given rotational speed and free stream speed, can serve as a bridge between helicopter-style Figure of Merit, and the usual definition of propulsive efficiency in fixed-wing aircraft forward flight. The best layout of propeller streamwise and vertical spacing, mindful of packaging constraints, is the one with the best ratio of thrust to power. Accordingly, the power required for the propeller pair (P_t) at each (d_y, d_z) and V_∞ is compared with the power required of two standalone single

propellers (P_s) generating the total thrust at the same respective V_∞ . The ratio of P_t and P_s denoted by P_t/P_s , represents the overall system power consumption of the tandem configuration when compared to a single propeller. To calculate P_t/P_s , the sum of the thrust (T_t) and power (P_s) for both tandem propellers is calculated at each V_∞ , then, using the single propeller performance data (Figure 4) between 67 and 117 RPS at the same V_∞ , the rotational speed of the single propeller that generates $T_s = T_t/2$ is found, along with the corresponding power consumption at this very rotational speed, P_s . The normalized power ratio P_t/P_s is then calculated and these steps are repeated for all test cases. $P_t/P_s = 1$ is the ideal, or zero interference. Any adverse interference between the propellers means $P_t/P_s > 1$, while $P_t/P_s < 1$, which is not physically likely, would be a favorable interaction that somehow exceeds the efficiency of propellers in isolation. Thus, the practical goal of a tandem propeller arrangement would be to drive P_t/P_s as close as possible to 1.0, subject to space constraints, across the range of advance ratios relevant to flight.

Figure 11 shows P_t/P_s vs. J and d_z at $\theta = 90^\circ$. The different colors in Figure 11 correspond to the vertical separation d_z , and the symbol shapes to different streamwise spacings, d_y . The variations in Figure 11 are larger between the various colors than the symbol shapes, or in other words, vertical spacing matters more than streamwise spacing, for $d_z \geq 0.4$. As the front propeller rotational speed is fixed, while the rear propeller rotational speed varies, the advance ratio is calculated with respect to the front propeller. In general, P_t/P_s remains ~ 1.0 at $J < 0.2$, as expected from Figure 7 to Figure 10. Between $0.2 < J < 0.4$, P_t/P_s increases for all cases, as a result of the reduction in C_T and increment in C_P for the rear propeller at this advance ratio range. At $J > 0.4$, changes in P_t/P_s with J become negligible for $d_z \geq 0.4$. This advance ratio boundary increased to $J = 0.6$ for $d_z = 0.2$, and even higher for the $d_z = 0.0$.

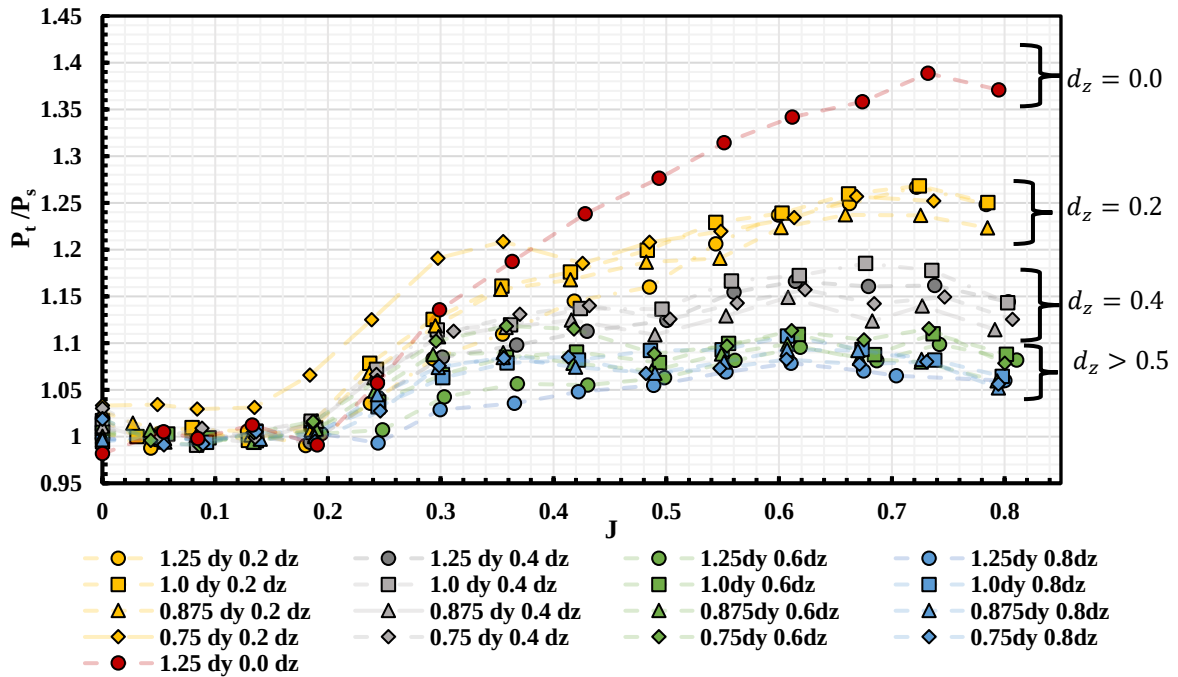


Figure 11. Ratio of P_t/P_s for all d_z cases at $\theta = 90^\circ$ vs. J .

Since at a lower d_z , the front propeller has a negative effect on the rear propeller, P_t/P_s becomes higher for lower d_z . A maximum P_t/P_s of 1.4 is observed at $d_z = 0$ at the highest advance ratios. P_t/P_s plateaus to 1.25 for $d_z = 0.2$ and 1.15 for $d_z = 0.4$. For $d_z > 0.5$, the difference is almost negligible between cases, with

a maximum $P_t/P_s \sim 1.07$ on average. As noted for thrust coefficient alone, so too for P_t/P_s , we find that $d_z = 0.4 - 0.5$ is a threshold, beyond which further vertical offset offers little aerodynamic benefit in tandem configurations. Within each d_z family (color), for large d_z , generally the lowest P_t/P_s (most efficient tandem configuration) is for the largest d_y (circular symbols). This is to be expected, as the lowest amount of inter-propeller interference should be where the propellers are spaced further apart. But the variations are small. For small d_z , the lowest P_t/P_s is actually not for the largest streamwise spacing, perhaps indicative of some favorable inter-propeller interaction. P_t/P_s however, is never below 1.0. The best propeller pairing possible is with the propellers infinitely far apart.

D) Tandem Propeller Configuration at $\theta = 75^\circ$

$\theta = 75^\circ$ is more representative of the multi-copter in cruise, where the vehicle has to pitch nose-down to generate a forward thrust component from the lifting-propellers. Similar to $\theta = 90^\circ$, the rear propeller is operated at a higher rotational speed to achieve pitching moment trim for the tandem propeller system.

And as with $\theta = 90^\circ$ in Figure 7 and Figure 8, the effect of the rear propeller on the front propeller is relatively small. An example of the front propeller C_T and C_P at $d_y = 1.25$ and various d_z is shown in Figure 12. At small d_z , C_T of the front propeller is slightly higher than the standalone propeller. When increasing d_z , a small decrement in C_T is observed at $J > 0.3$, such that in particular $d_z = 0.6$ has nearly the same C_T curve as the standalone propeller. For all d_z , the C_P is slightly higher than for the standalone case.

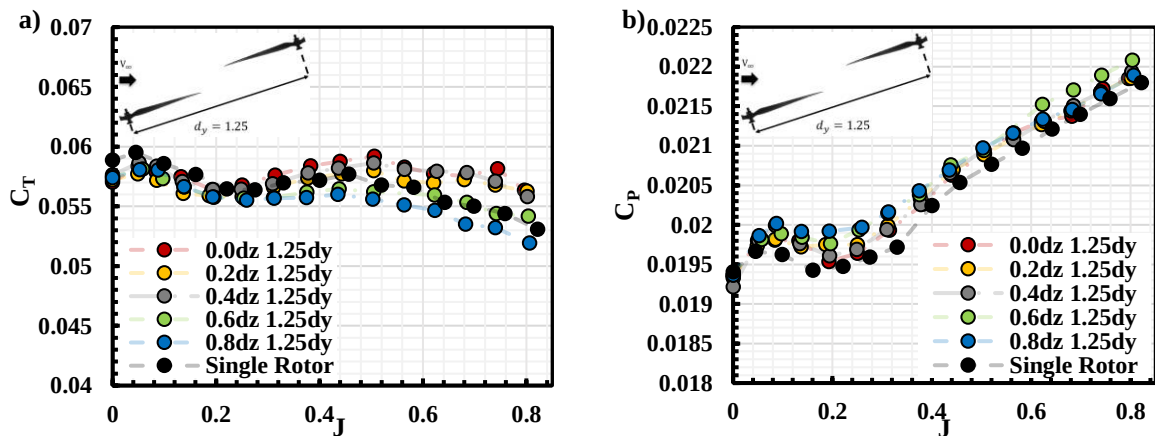


Figure 12. Front Propeller a) C_T and b) C_P vs. J at $\theta = 75^\circ$ and $d_y = 1.25$.

Figure 13 shows the rear propeller C_T at different J . Due to the propeller mounting interference, $d_y = 0.75$ cases are not tested in the experiment for $\theta = 75^\circ$ condition. At $J < 0.2$, C_T decreases with J for all cases, and the differences in C_T between cases are negligible. At $J > 0.2$, variation in C_T across d_z becomes significant. A lower d_z means more reduction in C_T , which is similar to the results for $\theta = 90^\circ$ in Figure 9. On the other hand, when reducing d_y to where the propeller starts to overlap, the reduction in C_T becomes less significant when compared to the non-overlapping cases, except for $d_z = 0.2$. Recall from the results in Figure 9, a similar trend is also observed as a result of placing the rear propeller above the front propeller.

C_P vs. J is shown in Figure 14. As compared to the $\theta = 90^\circ$ case in Figure 10, variation of C_P with both d_y and d_z is small, in all cases with a nearly linearly relationship with J , at the same slope. The supposition is that at this inclination angle, the influence of the front propeller on the rear is weaker than in the case of direct edgewise flight.

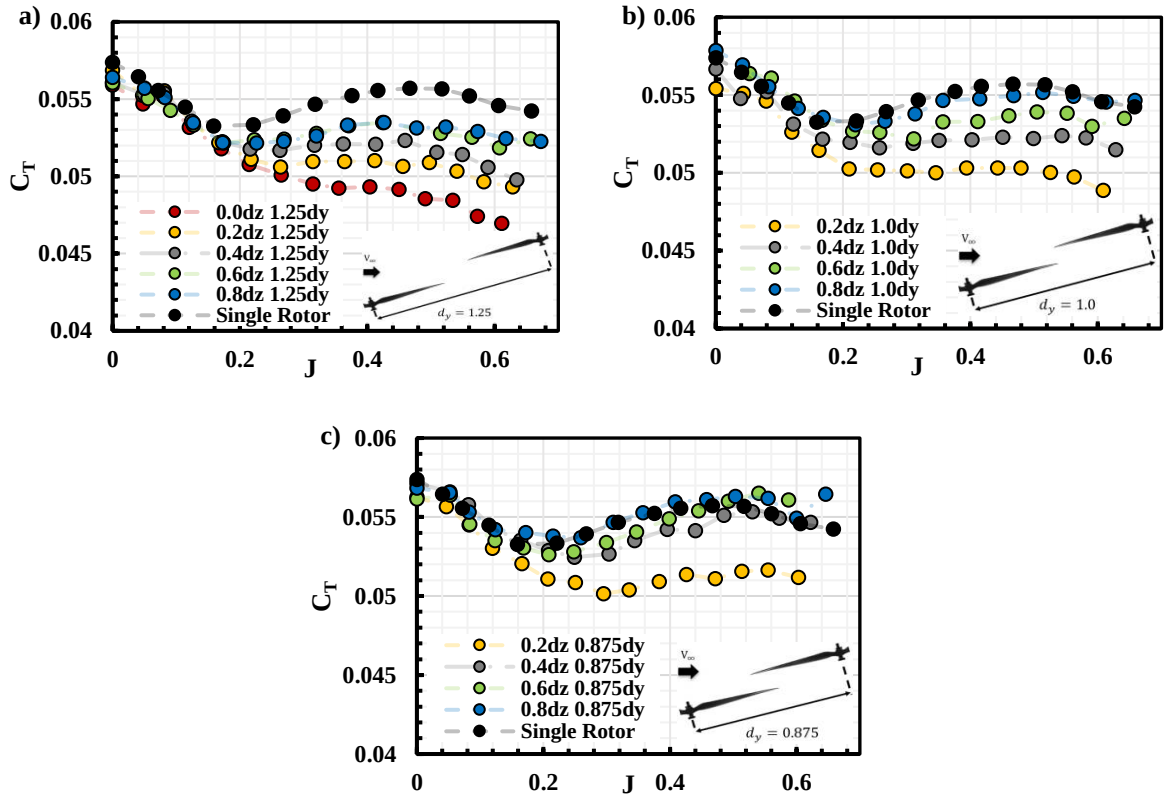


Figure 13. Rear propeller C_T vs. J at $\theta = 75^\circ$ for a) $d_y = 1.25$, b) $d_y = 1.0$ and c) $d_y = 0.875$.

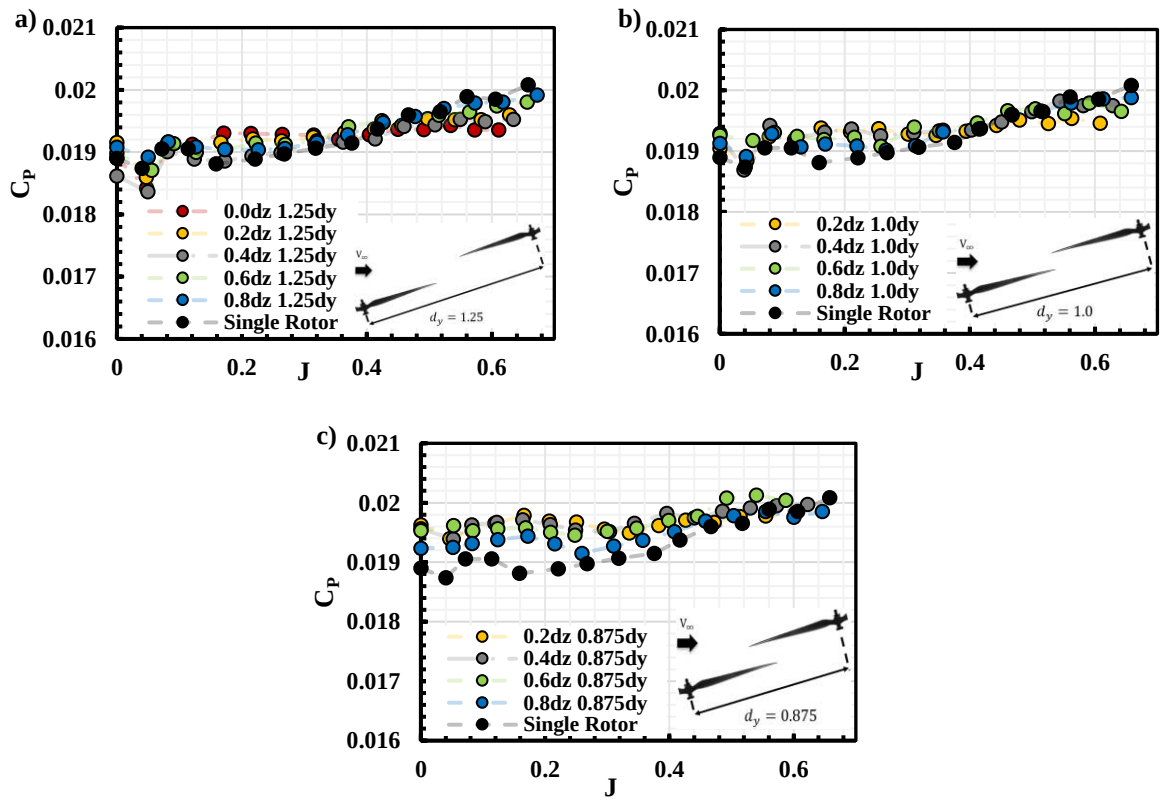


Figure 14. Rear propeller C_P vs. J at $\theta = 75^\circ$ for a) $d_y = 1.25$, b) $d_y = 1.0$ and c) $d_y = 0.875$.

P_t/P_s is calculated for the $\theta = 75^\circ$ case with the same method as in Figure 11, and shown in Figure 15. The results for two additional d_y cases at $d_z = 0.0$, $d_y = 1.5$ and $d_y = 1.75$ are given. As before,

variation in P_t/P_s due to vertical spacing (color of symbols in Figure 15) being larger than that due to streamwise spacing (symbols shape in Figure 15). Due to a less severe reduction in C_T (Figure 13) on the rear propeller, and a negligible changes in C_P (Figure 14) when compared to the $\theta = 90^\circ$ cases (Figure 9 and Figure 10), the overall magnitude for P_t/P_s is smaller than the $\theta = 90^\circ$ cases at the same propeller displacement, indicating that the interference effect of the tandem propeller at $\theta = 75^\circ$ is smaller than for $\theta = 90^\circ$. At $J < 0.15$, P_t/P_s remains about 1.0. For larger J , P_t/P_s increases, reaching its maximum between $0.35 < J < 0.5$, which is a common advance ratio range for a cruise condition in other studies [13, 17-19]. These trends are similar to the ones observed in Figure 11. A further increment in J results in a slight decrement in P_t/P_s .

The peak P_t/P_s for $d_z = 0$ is 1.1 at $d_y = 1.25$, and falls as d_y is increased. This stands to reason and is also observed in other studies [13,17]. Similar to the $\theta = 90^\circ$ cases, the peak P_t/P_s also falls as d_z is increased, with a maximum P_t/P_s of 1.07 for $d_z = 0.2$ and 1.05 for $d_z = 0.4$ across the range of advance ratio. Perhaps surprisingly, the lowest P_t/P_s at $d_z = 0.4$ is for $d_y = 0.875$, not for $d_y = 1.25$. At $d_z > 0.5$, $P_t/P_s \sim 1.0$.

It would be appealing if a given P_t/P_s at a given J could be reproduced through various combinations of d_y and d_z . A more compact drone would have smaller d_y at the expense of larger d_z . But because P_t/P_s varies with J above $J = 0.15$, the trade-space between larger overlap and larger vertical spacing is limited. An example from Figure 15 might be $d_y = 1.25$ and $d_z = 0.4$, vs. $d_y = 1.75$ and $d_z = 0.0$ at $J = 0.33$. But because d_z , up to 0.5, has a more powerful effect on P_t/P_s than does d_y , the implication would be to avoid low d_z if at all possible.

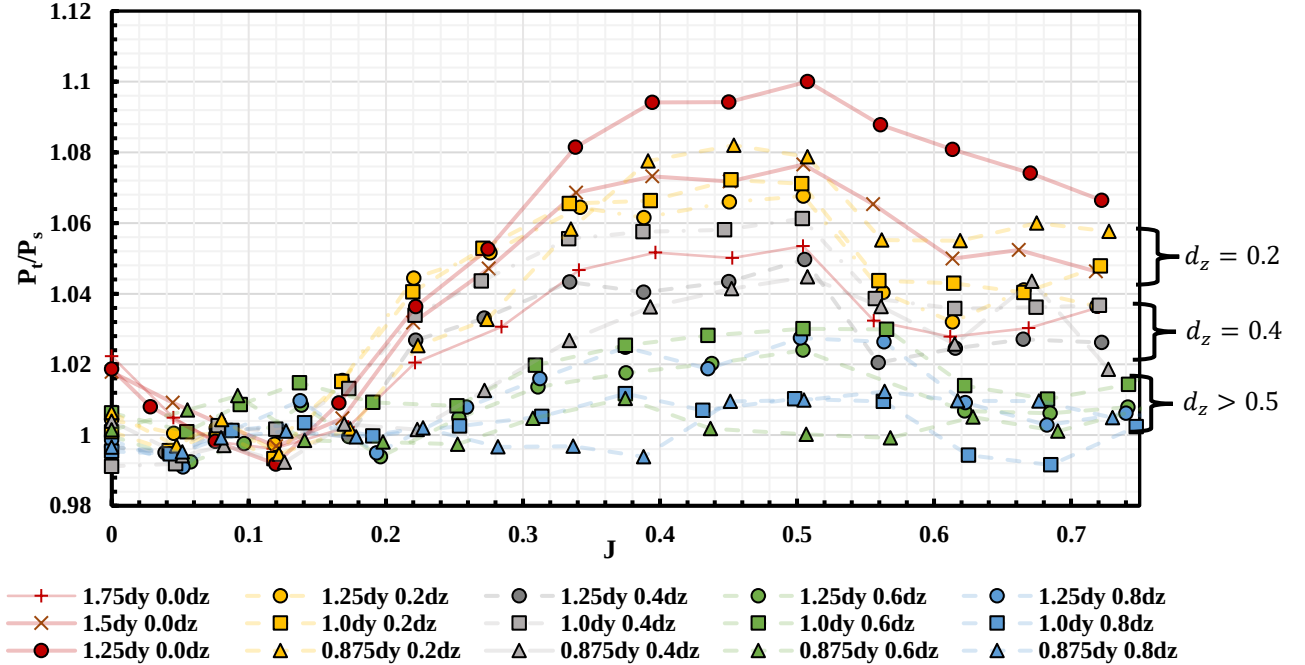


Figure 15. Ratio of P_t/P_s for all d_z cases at $\theta = 75^\circ$ vs. J .

The much smaller variation in P_t/P_s for $\theta = 75^\circ$ than for $\theta = 90^\circ$, suggests that spacing apart the propellers is of less importance if the drone tilts substantially to fly forward, than if it remains in purely edgewise flight, for example, if there is a separate source of horizontal thrust for forward propulsion. Thus, the drag penalty

of tilting forward, and the penalty of loss of vertical thrust component, have to be traded against the interference penalties captured in an increase in P_t/P_s .

V. Conclusions

Forces and moments were measured for a pair of propellers in tandem, with a front propeller upstream and a rear propeller downstream, across a range of advance ratios in an open-jet wind tunnel. The rear propeller was placed either coplanar with the front, or above it, with a parameter study of streamwise and vertical separation. The pair was either at 90-degree incidence with respect to the free stream, which is to say in purely edgewise flight; or at 75-degree incidence, the latter to model the cruise orientation of a conventional multi-copter. Propeller efficiency was reported as a quotient of the thrust coefficient and power coefficient. The efficiency of the tandem pair was referenced to that of propellers in isolation, by noting the power-required to produce the same thrust in each respective case and taking the ratio of these powers. The ideal for zero interference becomes a power-ratio of 1.0.

The power-ratio for a tandem pair, across the parameter space of vertical and streamwise separations, is a strong function of the advance ratio. Advance ratio of 0.15-0.2 is a threshold. Below that, the various combinations perform similarly, all with a power ratio of 1.01-1.05, which is to say, low interference, at least for the realized range of overlap. Higher overlap may be speculated to evince higher interference. Above advance ratio of ~0.2, the power ratio rises, to a peak of around 1.4 for pure edgewise flight, and a smaller peak of around 1.1 for flight at incidence. This suggests that the inclination of a drone to fly forward may improve propeller efficiency for compact configurations, as the advance ratio increases.

Increasing the vertical spacing between the front and rear propellers, starting from zero and going to half of the propeller diameter, results in a decrease in power ratio, or in other words, an increase in efficiency, though the threshold of 1.0 is never quite exceeded. Above a vertical spacing of one-half propeller diameter, there is no discernible further improvement, suggesting that further vertical spacing between propellers in a practical drone is not necessary, for the range of streamwise separations in this study. On the other hand, if the vertical separation between the propellers is zero, even a spacing three-fourths of a diameter between the respective propeller tips, is not sufficient to reach a power-ratio of 1.0, unless again the advance ratio is below 0.2.

VI. References

- [1] Leishman, J. G., "Principles of Helicopter Aerodynamics", Cambridge University Press, 2006.
- [2] Dingeldein, R. C., "Wind Tunnel Studies of the Performance of Multirotor Configurations," NACA TN 3236, August 1954.
- [3] Nagashima, T., O. H., and Sasaki, F., "Optimum Performance and Load Sharing of Coaxial Rotor in Hover," Journal of Japan Society for Aeronautics & Space Sciences, Vol. 26, No. 293, June 1978, pp. 325–333.
- [4] Lee, T. E., and Leishman, J. G., "Design and Performance of a Ducted Coaxial Rotor in Hover and Forward Flight," M. S. Thesis, University of Maryland, MD, October 2010.
- [5] Ramasamy, M., "Measurements Comparing Hover Performance of Single, Coaxial, Tandem, and Tilt-Rotor Configurations" AHS 69th Annual Forum, Phoenix, AZ, May 2013
- [6] Otsuka, H., Sasaki, D., and K. Nagatani, "Reduction of the head-up pitching moment of small quad-rotor unmanned aerial vehicles in uniform flow," Int. J. Micro Air Veh., pp. 1–21, 2017.

- [7] Otsuka, H., and Nagatani, K., "Pitching moment caused by rotor flow interference on a small quad-rotor unmanned aerial vehicle in a uniform flow," in Transactions of the Japan Society for Aeronautical and Space Sciences, 2018, vol. 61, no. 2, pp. 87–96.
- [8] Volocopter, VoloCity air taxi, Retrieved from: volocopter.com.
- [9] Fukushima, Y., and Oyama, A., "Aerodynamic interaction of overlapping rotors for Mars drone" AIAA SciTech 2022, January 2022, AIAA 2022-0731.
- [10] Felker, F. F., and J. S. Light, "Rotor / wing Aerodynamic Interactions in Hover," 1986.
- [11] Sweet, G. E., "Hovering measurements for twin-rotor configurations with and without overlap," Natl. Aeronaut. Sp. Adm., 1960.
- [12] Huston, J., "Wind-Tunnel Measurements of Performance, Blade Motions, and Blade Air Loads for Tandem-Rotor Configurations with and without Overlap," Hampton, Va, 1963.
- [13] Prudden, S. L., "Rotor Aerodynamic Interaction effect for multirotor unmanned aircraft system in forward flight " Thesis, RMIT University, Australia, September 2020
- [14] Tsaltas, J. D., "Trim Solutions of Multirotor Vehicles using a Fast Performance Prediction Method," Ryerson University, 2018.
- [15] Heyson, H. H., "An Evaluation of Linearized Vortex Theory As Applied To Single And Multiple Rotors Hovering In And Out Of Ground Effect," NASA TN D-43, September 1959
- [16] Tapscott, B. R. J., and Amer, K. B., "Studies of the speed stability of a tandem rotor helicopter in forward flight" 1956
- [17] Atte, A., Wylie, D., and Rauleder, J., "Experimental Evaluation of Multi-Rotor Aerodynamic Interactions", VFS 78th Annual Forum, May 2022.
- [18] McCrink, M., Seth, D., and Herz, S., "Flight Test Measurement of Quadrotor Performance at Varying Sideslip Angles" VFS 78th Annual Forum, May 2022.
- [19] Otsuka, H., and Nagatani, K., "Thrust Loss Saving Design of Overlapping Rotor Arrangement on Small Multirotor Unmanned Aerial Vehicles" 2016 IEEE ICRA, May 2016, IEEE-3242
- [20] ATI, F/T Mini-40 Sensor, Retrieved from: www.ati-ia.com
- [21] APC Propellers, Propeller technical data, Retrieved from: www.apcprop.com/
- [22] Pastor, G. R. et. al, "Numerical and Experimental Testing of a Coaxial Propeller for UAM Applications", AIAA Aviation 2022, June 2022. AIAA 2022-3683
- [23] Cai, J., Gunasekaran, S., and OL, M. V., "Sinusoidal Gust Response of RC Propellers at Different Incidence Angles" AIAA SciTech 2022, AIAA 2022-1536, Januray 2022.
- [24] McLemore, H. C., and Cannon, M. D., "Aerodynamic Investigation of a Four-blade Propeller Operating Through an Angle of Attack from 0 degree to 180 degree," NACA TN 3228, 1954.
- [25] Simmons, B. M., and Hatke, D. B., "Investigation of High Incidence Angle Propeller Aerodynamics for Subscale eVTOL Aircraft," NASA/TM-20210014010, 2021.
- [26] Cai, J., and Gunasekaran, S., "Propeller Ground and Ceiling Effect in Forward Flight", VFS 78th Annual Forum, May 2022.

[27] Cai, J., Gunasekaran, S., and OL, M. V., “Propeller Partial Ground and Ceiling Effect Prediction”, AIAA SciTech 2021, Virtual Event, January 2021, AIAA 2021-0614.